

Rental Property Price Prediction

Airbnb: Supervised Learning Regression

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Executive Summary

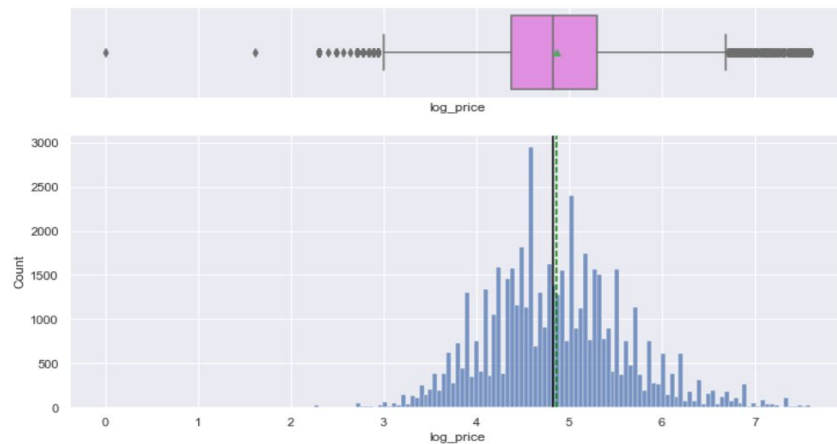
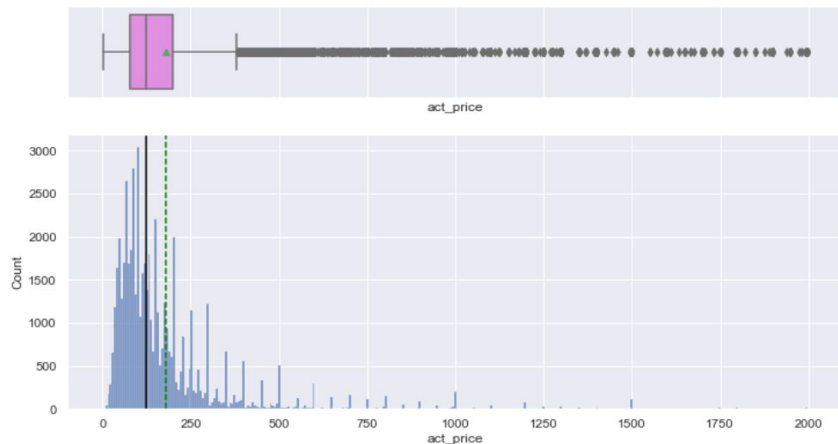
- With our linear regression model we have been able to capture ~51% of the variation in our data.
- Our final linear regression model has a MAPE of 8.3% on the test data, which means that we are able to predict within 8.3% of the rental price.
- Renting our rooms which can accommodate more people will fetch more revenue. Such rooms will generally be entire homes/apartments.
- The focus should be on providing a comfortable stay and better service to the customers to drive the review score ratings higher.
- We should explore data from other periods of time and about the ratings of services offered (dining, laundry, etc.). An analysis of the text reviews can also be conducted to gain an understanding of customer sentiment.
- We will also have to analyze the cost side of things before we can talk about profitability in the business. We should gather data regarding that.

Business Problem Overview and Solution Approach

- Airbnb is an online platform that allows people to rent short-term accommodation. This ranges from regular people with a spare bedroom to property management firms who lease multiple rentals. On the one side, Airbnb enables owners to list their space and earn rental money. On the other side, it provides travelers easy access to renting private homes.
- Airbnb receives commissions from two sources upon every booking, namely from the hosts and guests. For every booking, Airbnb charges the guest 6-12% of the booking fee. Moreover, Airbnb charges the host 3% for every successful transaction
- As a senior data scientist at Airbnb, you have to come up with a pricing model that can effectively predict the rent for accommodation and can help hosts, travelers, and also the business in devising profitable strategies.
- The objective is to explore and visualize the data, build a linear regression model to predict the prices of rental rooms, and generate a set of insights and recommendations that will help the business.

EDA Results

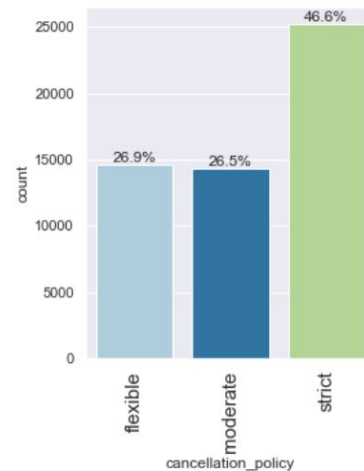
- The actual prices of the rental price of the property have skewed distribution and log prices of the rental price of the property are almost normally distributed



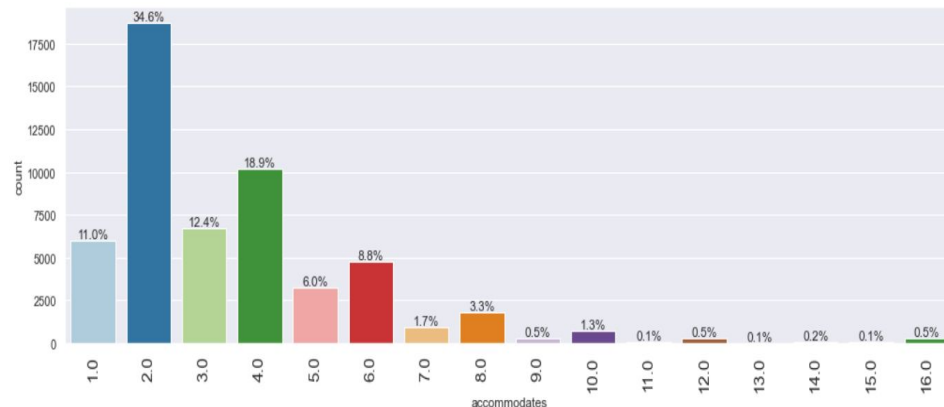
[Link to Appendix slide on data background check](#)

EDA Results

- Nearly 50% of the rooms have a strict cancellation policy.

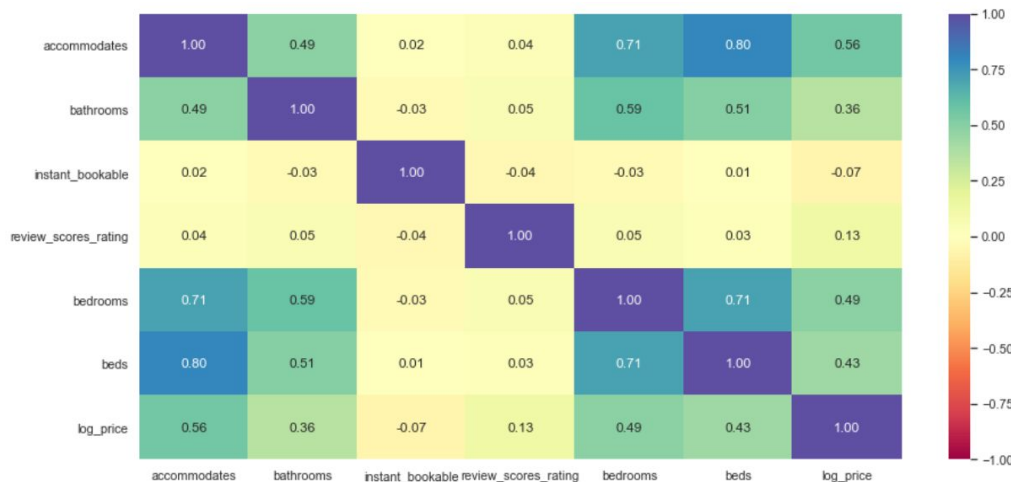


- Almost 75% of the rooms can accommodate 4 or less than 4 people.



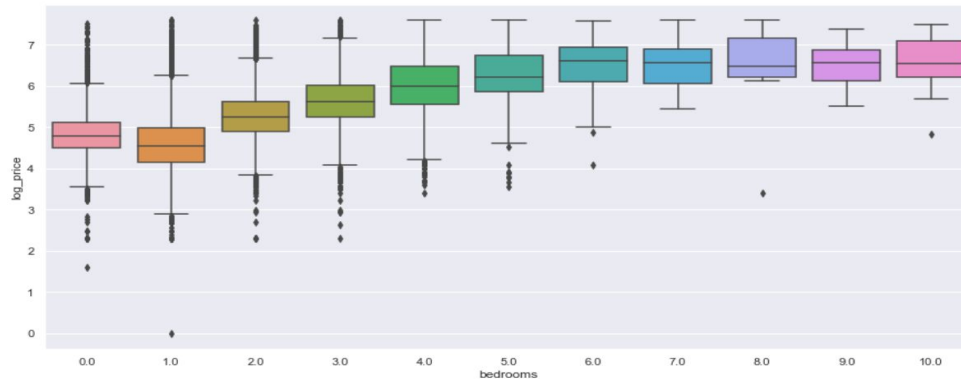
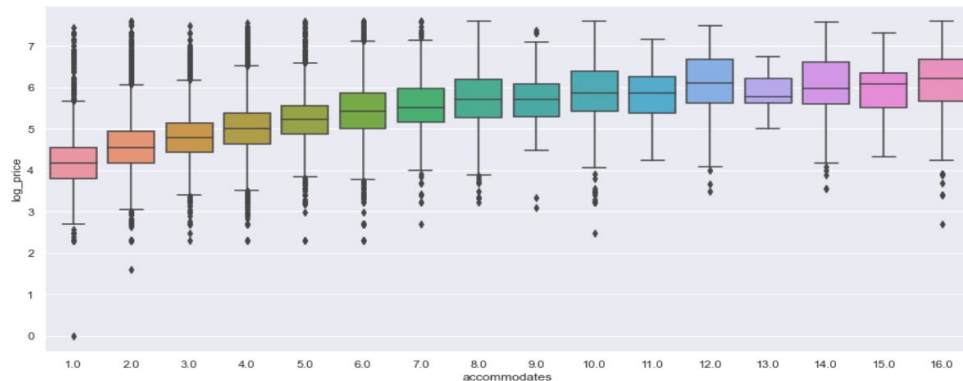
EDA Results

- The log price of a rental property is moderately positively correlated with the number of people it accommodates, bedrooms, and beds.
- Properties that can accommodate more people tend to have more beds.
- The number of bedrooms and beds in a rental property have a slightly high positive correlation.
- The size of a rental property, as indicated by the number of bedrooms and beds, is likely to affect its price.
- The correlation between the number of people a rental property accommodates and its price suggests that larger properties may be more expensive.



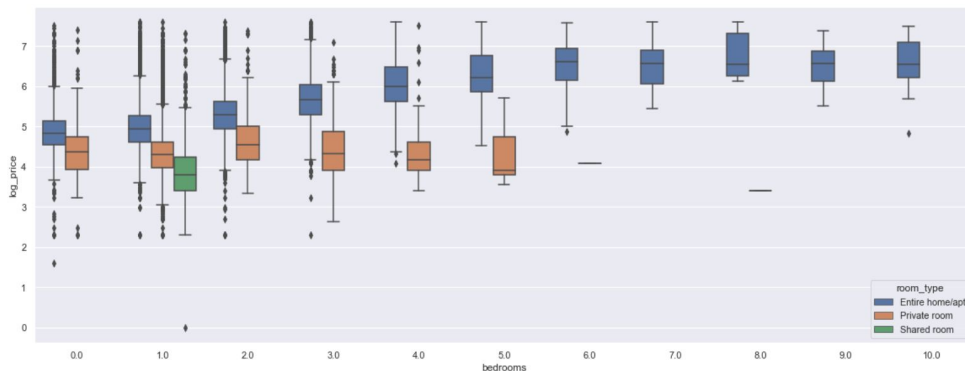
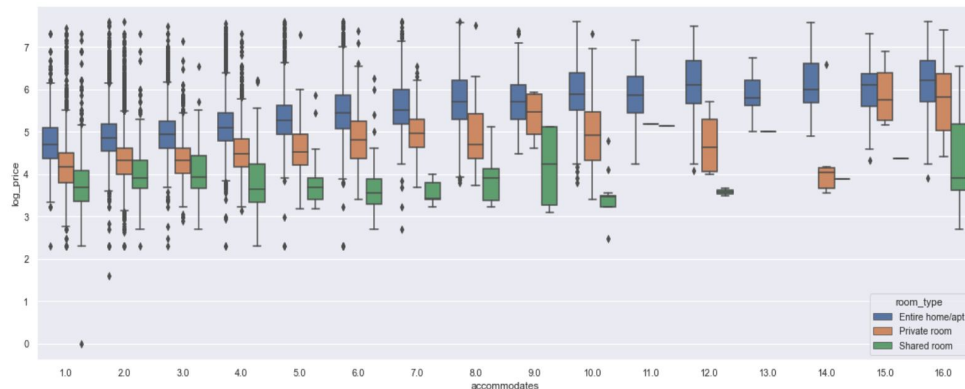
EDA Results

- Rooms allowing a higher accommodates have higher rental price.
- As the number of accommodates increases, rental price also increases.
- The rental price increases as the number of bedrooms increases.



EDA Results

- In general, the rental price for an entire home/apartment is more than the price of private rooms, which is again more than that of shared rooms.
- There are more upper outliers in terms of the rental price for entire rooms/apt which allow accommodation for 7 or fewer people. This indicates a higher rental price for some rooms.
- Shared rental rooms have only 1 bedroom.
- The rental price of entire homes or apartments is greater than that of private rooms, irrespective of the number of bedrooms.



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Data Preprocessing

- There are 19994 duplicate values in the data and it has been dropped
- The missing values in the data were treated as follows:
 - Missing values in numerical columns were imputed by the median values grouped by the room type
 - Missing values in non-numerical columns have been dropped
- There are outliers in a few columns and those are treated by flooring and capping.
- The actual price column is dropped as its distribution is right-skewed as there are many upper outliers so we used the log price column.
- The target variable log_price has a few values equal to 0 so we added a small positive value to them to avoid problems while checking the model performance.
- The columns "cleaning_fee", "instant_bookable" were boolean type column so we converted it into the integer type of column.

Model Performance Summary

- A linear regression model was built with the log price as the target variable
- The key metrics for assessing model performance have been summarized in the table below

Data	R^2	Adj. R^2	RMSE	MAE	MAPE
Train	0.509	0.509	0.515	0.400	8.405
Test	0.507	0.508	0.515	0.398	8.320

[Link to Appendix slide on model assumptions](#)

APPENDIX

Data Background and Contents

- The dataset contains information on 74111 property ids and their characteristics.
- The characteristics include Id, room type, accommodates, bathrooms, cancellation policy, cleaning_fee, instant bookable, review scores rating, bedrooms, beds and log_price and more.
- All numerical variables have outliers. Attributes accommodates, bedrooms, and beds have only upper outliers whereas reviews score have only lower outliers and log_price has both upper and lower outliers.



Happy Learning !

